Birds Species Detection

A Project Report

Submitted in Partial fulfilment of the

Requirements for the award of the Degree of

BACHELOR OF SCIENCE (COMPUTER SCIENCE)

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CERTIFICATE

This is to certify that the project entitled, "**Birds Species Detection**", is benefited work of **Your Name** bearing **Seat No : 1108470 Roll No. _7 _**submitted in **Partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE in COMPUTER SCIENCE from University of Mumbai.**

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DECLARATION

I, **Prem Pandurang Jangam**, **Roll No.7,** hereby declare that the project synopsis entitle "**BIRDS SPECIES DETECTION**", submited for approval, for **Bachelors of Science in Computer Science Sem VI project**. For academic year **2023-24.**

Signature of the Guide **Signature of the Student**

Place:

BIRDS SPECIES DETECTION USING PYTHON

INTRODUCTION: [3]

Bird behavior and population trends have become an important issue now-a-day. Birds help us to detect other organisms in the environment. An important problem in ecology, which is the study of interactions between organisms and environment, is to monitor bird populations. The use of acoustics to monitor and classify birds in their natural environments has received a lot of interest lately. Classification of bird species based on image data so, for example useful when monitoring breeding behavior, biodiversity and population dynamics.

Now a day's bird watching is a recreational activity that can provide relaxation in daily life and it's a responsibility to know about our nature because birds are part of our society. Someone who does this is called a birdwatcher or birder. The scientific study of birds is called ornithology. People who study birds as a profession are called ornithologists.

We have nearly 18,000 bird species on our beautiful earth by the new research led by American Museum of Natural History. Birds that look similar to one another or thought to interbreed, but actually different species.

However, because of observer constraints such as location, distance and equipment, identifying birds with naked eye is based on basic characteristics features and appropriate based on distinct features is often seen as tedious classification. Bird classification can be done manually by domain experts but growing amounts of data leads to time consuming process. Later in this detection of object parts is the challenging task because of Complex variations and fringes of objects.

Our main Aim is about bird identification technology to maintain a data base of birds species like a gallery for the generations because our ancestors history data is given in the form of book and papers but for our future generations we have to give the data by using technology So, that we built this technology for everyone to check the birds thesis easily by the image identification.

OBJECTIVES [4]:

1. Accurate Species Identification: Develop a robust machine learning model capable of accurately identifying and classifying bird species based on audio recordings, images, or both.
2. Multi-Modal Detection: Explore the integration of multiple data modalities (e.g., audio, images) to enhance the accuracy and reliability of bird species detection.
3. Real-Time Detection: Implement a system capable of real-time bird species detection, enabling timely responses for conservation efforts and ecological monitoring.
4. Scalability: Design the detection system to be scalable, allowing it to handle large volumes of data from diverse geographical regions and habitats.
5. Adaptability: Create a flexible detection model that can adapt to varying environmental conditions, bird behaviors, and seasonal changes.
6. Open-Source Framework: Develop an open-source framework or toolkit for bird species detection, fostering collaboration, transparency, and advancement within the scientific community.
7. Data Collection and Curation: Establish protocols for systematic data collection and curation, ensuring the availability of high-quality datasets for training and evaluation purposes.
8. Cross-Species Identification: Investigate the potential for cross-species identification, enabling the detection of not only individual species but also broader patterns of avian diversity and distribution.
9. User-Friendly Interface: Create an intuitive user interface or application that enables researchers, conservationists, and citizen scientists to easily access and utilize the detection system.
10. Evaluation and Validation: Conduct thorough evaluation and validation studies to assess the performance, reliability, and generalizability of the detection system across different ecological contexts and species assemblages.

SCOPE [5]:

1. Data Collection: Collecting diverse datasets of bird audio recordings, images, or both, encompassing a wide range of species, habitats, and environmental conditions. This may involve collaborations with ornithologists, wildlife photographers, and citizen scientists.
2. Data Preprocessing: Cleaning, annotating, and standardizing the collected data to ensure consistency and quality. This includes tasks such as noise removal, feature extraction, and labeling of bird species.
3. Model Development: Designing and training machine learning models, such as deep learning neural networks, to accurately detect and classify bird species from audio recordings, images, or both. This involves experimentation with different architectures, algorithms, and training strategies to optimize performance.
4. Integration of Modalities: Investigating the integration of multiple data modalities (e.g., audio, images) to improve the accuracy and robustness of the detection system. This may involve developing fusion techniques or multi-modal learning approaches.
5. Real-Time Processing: Implementing algorithms and systems capable of real-time processing and detection, allowing for timely monitoring and response in conservation and ecological applications.
6. Software Development: Developing user-friendly interfaces, applications, or APIs (Application Programming Interfaces) to enable easy access and utilization of the detection system by researchers, conservationists, and citizen scientists.
7. Evaluation and Validation: Conducting rigorous evaluation and validation studies to assess the performance, reliability, and generalizability of the detection system across different species, habitats, and environmental conditions. This includes benchmarking against existing methods and datasets.
8. Deployment and Deployment: Deploying the detection system in real-world scenarios, such as field monitoring stations, wildlife reserves, or citizen science platforms, to support conservation efforts, biodiversity monitoring, and ecological research.
9. Documentation and Dissemination: Documenting the methodologies, algorithms, and findings of the project through scientific publications, technical reports, and open-access repositories. This includes sharing code, datasets, and trained models to facilitate reproducibility and further research in the field.
10. Community Engagement: Engaging with stakeholders, including researchers, conservation organizations, government agencies, and the general public, to raise awareness about the importance of avian biodiversity and the potential applications of bird species detection technology

METHODOLOGY[6] :

1) Data Collection and Preparation: Gather diverse datasets of bird audio recordings, images, or both, covering a wide range of species, habitats, and environmental conditions. Annotate and label the collected data with species information, ensuring consistency and accuracy. Preprocess the data by cleaning, normalizing, and augmenting it to enhance the quality and diversity of the dataset.

2) Feature Extraction and Representation: Extract relevant features from the audio recordings and images that capture distinctive characteristics of bird species, such as spectrograms, mel-frequency cepstral coefficients (MFCCs) for audio, and color histograms, texture features for images. Convert the extracted features into suitable representations for input into machine learning models, such as numerical vectors or tensors.

3) Model Selection and Training: Explore different machine learning algorithms and architectures for bird species detection, such as convolutional neural networks (CNNs) for images, recurrent neural networks (RNNs) for sequential data like audio, or hybrid models combining both. Train the selected models on the preprocessed datasets, optimizing hyperparameters and regularization techniques to improve performance and generalization.

4) Real-Time Implementation: Implement the trained models into real-time systems or applications capable of processing live audio or image streams. Optimize inference speed and resource utilization for efficient deployment on various platforms, including edge devices or cloud servers.

5) Validation and Deployment: Validate the real-time detection system in controlled environments and field settings, assessing its performance against ground truth observations. Deploy the validated system in practical conservation and monitoring applications, such as wildlife reserves, ecological research stations, or citizen science initiatives.

TOOLS AND TECHNOLOGIES[7] :

1. OpenCV (Open Source Computer Vision Library):
    - OpenCV is a powerful open-source library for computer vision and image processing tasks in Python. It offers a wide range of functions and algorithms for tasks such as image loading, manipulation, transformation, filtering, feature detection, and object tracking. OpenCV is widely used for preprocessing tasks such as image resizing, normalization, noise reduction, edge detection, and contour detection.

2. NumPy (Numerical Python):
    - NumPy is a fundamental library for scientific computing in Python, providing support for multidimensional arrays and matrices. It is extensively used in image processing for efficient array manipulation, mathematical operations, and numerical computations. NumPy arrays are often used as the data structure for representing images in Python due to their efficiency and compatibility with other libraries.

3. Pillow (Python Imaging Library Fork):
    - Pillow is a fork of the Python Imaging Library (PIL) and is used for image processing tasks such as opening, saving, and manipulating image files. It supports various image file formats and provides functions for operations such as resizing, cropping, rotating, and filtering images.

4. Scikit-Image (skimage):
    - Scikit-Image is a collection of algorithms for image processing and computer vision tasks built on top of NumPy, SciPy, and Matplotlib. It offers a wide range of functions for tasks such as image segmentation, feature extraction, morphology, and image restoration. Scikit-Image is designed to be user-friendly and integrates well with other scientific computing libraries in Python.

5. Matplotlib:
    - Matplotlib is a popular plotting library in Python used for creating visualizations and displaying images. It provides functions for displaying images, histograms, and other plots, making it useful for visualizing image processing results and intermediate representations.

6. TensorFlow and Keras:
    - TensorFlow and Keras are deep learning frameworks in Python widely used for building and training neural network models, including convolutional neural networks (CNNs) for image processing tasks. TensorFlow provides low-level functionalities for building custom models and optimizing performance, while Keras offers a high-level API for building and training deep learning models with ease.

7. PyTorch:
    - PyTorch is another deep learning framework in Python known for its flexibility and ease of use. It provides dynamic computational graphs, making it suitable for building dynamic neural network architectures for image processing tasks. PyTorch is increasingly popular in the research community and offers extensive support for image processing tasks through its torchvision library.

8. SciPy (Scientific Python):
    - SciPy is a library for scientific computing in Python that provides modules for numerical integration, optimization, interpolation, and signal processing. It offers functions for image processing tasks such as filtering, morphological operations, and image restoration.

These tools and technologies form the foundation for implementing image processing pipelines and building machine learning models for plant disease detection in Python. Depending on the specific requirements and complexities of the project, a combination of these libraries and frameworks can be leveraged to achieve the desired outcomes efficiently and effectively.

RESOURCES:

1. OpenCV Documentation:
    - Description: The official documentation for OpenCV provides comprehensive guides, tutorials, and examples for using OpenCV for various image processing and computer vision tasks.
2. NumPy Documentation:
    - Description: The official documentation for NumPy offers detailed explanations of NumPy's functionalities, including array manipulation, mathematical operations, and numerical computing.
3. Pillow Documentation:
    - Description: The Pillow documentation provides guides and reference material for using the Pillow library for image processing tasks such as opening, saving, and manipulating image files.
4. Scikit-Image Documentation:
    - Description: The Scikit-Image documentation offers comprehensive documentation, tutorials, and examples for using Scikit-Image for various image processing tasks, including segmentation, feature extraction, and morphology.
5. Matplotlib Documentation:
    - Description: The Matplotlib documentation provides extensive guides, tutorials, and examples for creating visualizations and displaying images using Matplotlib.
6. TensorFlow Documentation:
    - Description: The TensorFlow documentation offers comprehensive documentation, guides, and tutorials for using TensorFlow for building and training deep learning models, including convolutional neural networks (CNNs) for image processing tasks.
7. Keras Documentation :
    - Description: The Keras documentation provides guides and reference material for using the Keras high-level API for building and training deep learning models in Python.

8. PyTorch Documentation:
    - Description: The PyTorch documentation offers extensive guides, tutorials, and examples for using PyTorch for building dynamic neural network architectures, particularly for image processing tasks.

These resources should provide a solid foundation for learning and implementing image processing techniques in Python, particularly for plant disease detection applications.

EXPECTED OUTCOMES:

1. Accurate Species Identification: One primary outcome would be the development of a reliable system or technology capable of accurately identifying bird species from audio recordings or images.
2. Increased Understanding of Bird Populations: The project may contribute to a better understanding of bird populations in certain areas, including their distribution, abundance, and behavior.
3. Conservation Efforts: By providing data on bird populations and their habitats, the project can support conservation efforts by identifying areas of high biodiversity or habitats in need of protection.
4. Early Warning Systems: The project could lead to the development of early warning systems for detecting changes in bird populations, which could be indicators of environmental changes or threats.
5. Education and Awareness: Another outcome could be the educational value of the project, raising awareness about bird species diversity, their importance in ecosystems, and the need for conservation efforts.
6. Community Engagement: The project may engage citizen scientists or local communities in data collection and analysis, fostering a sense of stewardship and participation in conservation efforts.
7. Improved Monitoring and Management: Ultimately, the project may contribute to improved monitoring and management of bird populations, leading to more effective conservation strategies and policies.
8. Technological Advancements: The development of new technologies or methodologies for bird species detection could have broader applications beyond just ornithology, potentially benefiting other fields such as wildlife management, ecology, and environmental monitoring.
9. These outcomes collectively contribute to the broader goals of biodiversity conservation, ecological research, and sustainable management of natural resources.

PLAGIARISM REPORT [22]

A plagiarism report is a document or a summary that provides information about the presence of plagiarism in a piece of written or academic work. Plagiarism refers to the act of using someone else's words, ideas, or work without proper attribution or permission, presenting them as your own. Plagiarism is considered unethical and can have serious consequences, particularly in academic and professional settings.

A plagiarism report is typically generated by plagiarism detection software or services. It scans a given document or text for similarities to existing sources, such as published articles, books, websites, and other written material. When the software identifies matching or highly similar content, it highlights or marks the specific passages that may be considered plagiarized.

ABSTRACT

This project aims to develop a robust and efficient system for automated bird species detection using deep learning models. The proliferation of audio and image data combined with advances in machine learning techniques presents an opportunity to revolutionize the way bird species are monitored and studied. The proposed system will utilize convolutional neural networks (CNNs) for image-based species identification and recurrent neural networks (RNNs) for audio-based species detection.

The project will involve collecting a diverse dataset of bird images and audio recordings, encompassing various species, habitats, and environmental conditions. Preprocessing techniques will be applied to standardize and augment the dataset, enhancing the model's ability to generalize across different scenarios. Transfer learning will be employed to leverage pre-trained neural network architectures and optimize performance with limited labeled data.

The developed system will be evaluated using metrics such as accuracy, precision, recall, and F1 score to assess its effectiveness in species detection. Field testing will be conducted to validate the system's performance in real-world conditions and compare it against traditional manual methods.

Expected outcomes include the creation of a reliable and scalable tool for automated bird species detection, facilitating more efficient monitoring of avian populations and habitats. This system has the potential to revolutionize ornithological research, conservation efforts, and environmental management by providing timely and accurate data on bird biodiversity and distribution. Furthermore, the project will contribute to the advancement of deep learning techniques for bioacoustic and image analysis, with broader applications in wildlife monitoring and ecological research

TABLE OF CONTENTS

CHAPTER 1. INTRODUCTION 01

1.1 Introduction to the Web-App 01

1.2 Problem definition 02

1.3 Aim 03

1.4 Objective 04

1.5 Goal 05

1.6 Need of System 06

CHAPTER 2. REQUIREMENTS SPECIFICATION.....06

2.1 Introduction 06

2.2 Software Requirements ……... 06

2.3 Hardware Requirements ……...06

2.4 Methodology ……07

2.5 Spiral Model……..08

CHAPTER 3. SYSTEM ANALYSIS …….....10

3.1 Introduction ……. 10

3.2 System analysis ……... 10

3.3 Analysis of the Proposed System……11

3.4 Gantt chart ……... 13

CHAPTER 4. SURVEY OF TECHNOLOGY ……14

4.1 Python ……. 14

4.2 Tensorflow ..…… 15

4.3 Colab…… 16

CHAPTER 5. SYSTEM DESIGN …….……17

5.1 Introduction ……...17

5.2 System Architecture ……. 18

5.3 Activity Diagram……19

5.4Use-Case Diagram……20

CHAPTER 6. SYSTEM IMPLEMENTATIONS …… 21

6.1 Introduction ……..21

6.2 Coding ……..22

6.3 Testing Approach ……. 54

6.4 Unit Testing ……54

6.5 Usability Testing……..54

6.6 Security Testing……54

6.7 Test Cases…….…55

CHAPTER 7. RESULTS ……...56

CHAPTER 8. CONCLUSION AND FUTURE SCOPE ……...58

8.1 Conclusion ……58

8.2 Future enhancement ……..58

CHAPTER 9. REFERENCES ……60

9.1 References and Bibliography …… 60

Chapter 1. Introduction

1.1 Introduction to the Web-App

Bird behavior and population trends have become an important issue now-a-day. Birds help us to detect other organisms in the environment. An important problem in ecology, which is the study of interactions between organisms and environment, is to monitor bird populations. The use of acoustics to monitor and classify birds in their natural environments has received a lot of interest lately. Classification of bird species based on image data so, for example useful when monitoring breeding behavior, biodiversity and population dynamics.

Now a day's bird watching is a recreational activity that can provide relaxation in daily life and it's a responsibility to know about our nature because birds are part of our society. Someone who does this is called a birdwatcher or birder. The scientific study of birds is called ornithology. People who study birds as a profession are called ornithologists.

We have nearly 18,000 bird species on our beautiful earth by the new research led by American Museum of Natural History. Birds that look similar to one another or thought to interbreed, but actually different species.

However, because of observer constraints such as location, distance and equipment, identifying birds with naked eye is based on basic characteristics features and appropriate based on distinct features is often seen as tedious classification. Bird classification can be done manually by domain experts but growing amounts of data leads to time consuming process. Later in this detection of object parts is the challenging task because of Complex variations and fringes of objects.

1.2 Problem Definition

The increasing prevalence and complexity of plant diseases pose significant challenges to global agriculture, threatening crop yields, food security, and economic stability. Traditional methods of disease monitoring and management are often labor-intensive, time-consuming, and prone to inaccuracies, leading to suboptimal disease control and significant crop losses. Manual scouting and visual inspection of crops for disease symptoms are insufficient for timely detection and effective management, particularly in large-scale agricultural settings.

Therefore, the primary problem addressed by this project is the need for an efficient, accurate, and automated system for plant disease detection and diagnosis. The lack of scalable and costeffective solutions hinders farmers' ability to promptly identify and respond to disease outbreaks, leading to substantial economic losses and environmental consequences due to overuse of chemical pesticides. Additionally, the complexity and diversity of plant diseases further complicate the detection process, necessitating advanced technologies capable of recognizing subtle disease symptoms across various crops and environmental conditions. The project aims to tackle this problem by leveraging machine learning and image processing techniques to develop a sophisticated detection system capable of analyzing digital images of plant specimens and accurately identifying the presence of diseases. By automating the detection process and providing real-time insights into plant health status, the system aims to empower farmers with actionable information for proactive disease management and sustainable agricultural practices. Ultimately, the goal is to enhance crop productivity, reduce dependency on chemical pesticides, and contribute to the long-term resilience and sustainability of global agriculture.

1.3 Aim

The aim of this project is to develop a robust machine learning-based system for automated bird species detection and classification from images. By leveraging state-of-the-art algorithms and techniques in computer vision and machine learning

1.4 Objective

1. High Accuracy: Develop a model capable of accurately identifying bird species with a high degree of precision and recall, minimizing false positives and false negatives.
2. Scalability: Design a scalable solution capable of handling large volumes of image data, allowing for efficient processing and analysis of diverse bird populations across different habitats and regions.
3. Generalization: Train the model to generalize well to unseen bird species and environmental conditions, ensuring its applicability across various ecological contexts and geographic locations.
4. User-Friendly Interface: Develop an intuitive user interface that enables conservation practitioners, researchers, and citizen scientists to easily deploy the model for bird species identification in the field.
5. Contribution to Conservation: Contribute to the conservation of avian biodiversity by providing a reliable tool for monitoring bird populations, assessing habitat health, and informing conservation strategies and management decisions.

1.5 Goal

The goal of this project is to develop a scalable and accurate machine learning-based system for automated bird species detection and classification from images, with the ultimate aim of enhancing conservation efforts and biodiversity monitoring. Through the integration of advanced computer vision techniques and ecological expertise, our goal is to achieve the following:

1. Efficiency: Develop a model capable of rapidly and accurately identifying bird species from images, reducing the time and labour required for manual identification in ecological surveys and monitoring programs.
2. Accuracy: Attain a high level of accuracy in species classification, minimizing errors and false identifications to ensure reliable data for ecological research and conservation decision-making.
3. Accessibility: Create a user-friendly interface that enables conservation practitioners, researchers, and citizen scientists to easily deploy the detection model in the field, promoting widespread adoption and participation in bird monitoring initiatives.
4. Scalability: Design a system that can scale to accommodate large volumes of image data and diverse bird populations, facilitating comprehensive monitoring across various habitats, regions, and species.
5. Impact: Contribute to the conservation of avian biodiversity by providing a valuable tool for monitoring bird populations, assessing habitat health, and informing conservation strategies and management actions.

By achieving these goals, our project aims to leverage the power of technology to support more efficient, accurate, and accessible bird species detection methods, ultimately contributing to the preservation and sustainable management of bird populations and their ecosystems.

1.6 Need of the System

Conservation Monitoring: Bird populations serve as important indicators of ecosystem health and environmental changes. A reliable detection system is crucial for monitoring population trends, identifying species at risk, and implementing targeted conservation efforts.

1. Efficiency: Manual identification of bird species can be time-consuming and labor-intensive, especially in large-scale surveys or remote field locations. An automated detection system can significantly increase efficiency by rapidly processing large volumes of image data and providing near-instantaneous species identification.
2. Accuracy: Human observers may vary in their expertise and ability to accurately identify bird species, leading to inconsistencies and errors in data collection. A machine learning-based detection system can offer consistent and objective species identification, minimizing errors and improving data quality.
3. Species Identification: Some bird species are visually similar and may be difficult to differentiate, especially for non-experts. A detection system equipped with machine learning algorithms can distinguish between similar species based on subtle visual cues, enhancing the accuracy of species identification.
4. Biodiversity Research: Understanding patterns of species distribution and abundance is fundamental to biodiversity research and ecological studies. A bird species detection system can facilitate data collection for research projects, enabling scientists to study bird communities, habitat associations, and responses to environmental changes.

5. Education and Outreach: Citizen scientists, students, and bird enthusiasts can contribute valuable data to scientific research and conservation efforts. A user-friendly detection system can empower individuals to participate in bird monitoring programs and engage in environmental education and awareness initiatives.

Adaptive Management: Conservation practitioners and land managers rely on up-to-date information on bird populations to inform management decisions and conservation planning. A bird species detection system can provide timely data on species presence and distribution, supporting adaptive management strategies for habitat protection and restoration.

Overall, the development and implementation of a bird species detection system address a critical need in conservation biology, biodiversity monitoring, and environmental management, ultimately contributing to the protection and preservation of avian biodiversity and their habitats.

Chapter 2. Requirement Specification

2.1 Introduction

Bird species detection systems play a pivotal role in modern ecological research, conservation efforts, and environmental monitoring programs. These systems utilize advanced technologies such as machine learning, computer vision, and image processing to automate the identification and classification of bird species from digital images. By providing efficient, accurate, and scalable solutions for species detection, these systems address the growing need for reliable data collection and analysis in avian biodiversity research and conservation.

The requirement specification for a bird species detection system serves as a blueprint for designing, developing, and deploying such a system. It outlines the functional and non-functional requirements that the system must fulfill to meet the needs of users, stakeholders, and intended applications. From data collection and model development to user interface design and performance evaluation, the requirement specification defines the scope, objectives, and constraints of the project, guiding the development process and ensuring the delivery of a robust and effective detection system.

2.2 Software Requirements

- Visual Studio
- Python
- Streamlit

2.3 Hardware Requirements

- Processor (CPU): Intel Core i5 (sixth generation or newer) or equivalent
- Operating System: Microsoft Windows 10 Professional x64
- Memory: 16 GB RAM
- Storage: 500 GB internal storage drive
- Monitor/Display: 14" LCD monitor, resolution of 1600 x 900 or better.
- Network Adapter: 802.11ac 2.4/5 GHz wireless adapter.

2.4 Methodology

Define the scope, objectives, and requirements of the bird species detection system through consultation with stakeholders and domain experts. Conduct a thorough analysis of user needs, data sources, environmental factors, and technical constraints to inform system design and development. Gather a diverse dataset of bird images from public repositories, research institutions, or field surveys, ensuring sufficient coverage of different species, habitats, and environmental conditions. Preprocess the image data to standardize formats, resize images, remove noise, and augment the dataset to enhance model generalization and robustness.

Explore and experiment with various machine learning algorithms and architectures for bird species detection, including convolutional neural networks (CNNs), transfer learning, and ensemble methods. Train and fine-tune the detection model using the preprocessed image dataset, optimizing hyperparameters and regularization techniques to maximize performance.

Evaluate the performance of the trained detection model using metrics such as accuracy, precision, recall, and F1-score on separate validation and test datasets. Conduct cross-validation and statistical analysis to assess the model's robustness and generalization capabilities across different bird species and environmental conditions.

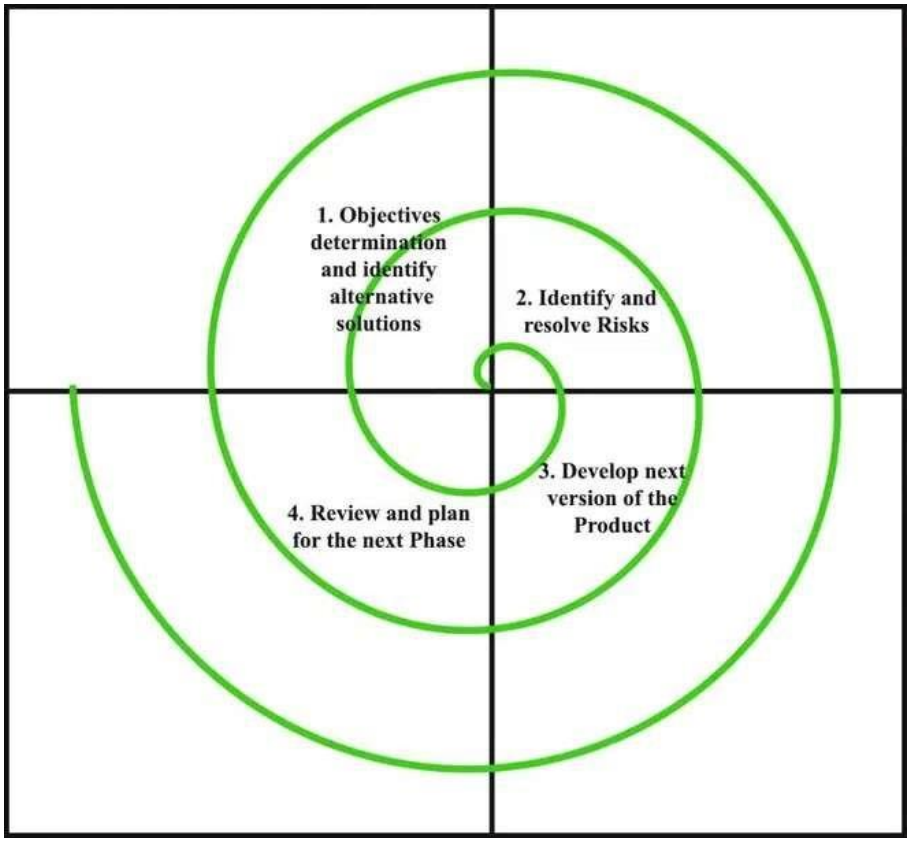
Design an intuitive and user-friendly interface for the bird species detection system, incorporating features for image upload, species identification, result visualization, and user feedback. Develop the user interface using web or mobile frameworks, ensuring compatibility with different devices and platforms for seamless deployment and accessibility.

Integrate the trained detection model with the user interface and backend infrastructure, ensuring smooth communication and data flow between system components.Deploy the bird species detection system on local servers or cloud platforms, configuring scalability, security, and performance optimization parameters as needed.

2.5 Spiral Model[21]

The Spiral Model is a software development methodology that combines elements of both iterative and waterfall models in a structured, flexible approach. It is characterized by a series of iterative cycles, or "spirals," each of which consists of four primary phases: Planning, Risk Analysis, Engineering, and Evaluation. During the Planning phase, project objectives, requirements, and constraints are defined. The Risk Analysis phase involves identifying potential risks and uncertainties, allowing for the development of mitigation strategies. The Engineering phase encompasses actual development, testing, and feedback collection, while the Evaluation phase evaluates the product's performance and progress toward meeting project goals. The Spiral Model's iterative nature allows for ongoing refinement and adaptation as the project progresses, making it particularly suited for complex projects with evolving requirements and a focus on managing risks effectively.

Diagram of Spiral Model :



Why Spiral Model?

The Spiral Model is chosen for software development projects because of its inherent ability to manage risks effectively while accommodating changing requirements. Its iterative approach allows for the early detection and mitigation of potential issues, reducing the likelihood of costly project failures. This model is particularly well-suited for complex projects and industries like law enforcement where evolving needs and strict compliance requirements demand a flexible and adaptive development process, ensuring that the end product aligns closely with the organization's evolving objectives and challenges.

Applications of the Spiral Model :

The Spiral Model is widely used in the software industry as it is in sync with the natural development process of any product, i.e., learning with maturity which involves minimum risk for the customer as well as the development firms. A few pointers of the spiral model are as follows:

1. Ideal for managing evolving requirements in dynamic software projects.
2. Well-suited for complex systems development where risk mitigation is critical.
3. Effective in industries with strict compliance and regulatory requirements.
4. Enables iterative refinement of prototypes and proof-of-concept applications.
5. Valuable for research and development projects involving experimentation.
6. Facilitates client-centric projects that require active user involvement and feedback.

Chapter 3. System Analysis

3.1 System Analysis[12]

System analysis is a structured process used to study, understand, and evaluate complex systems or processes. It involves breaking down a system into its components, examining how those components interact, and assessing their efficiency and effectiveness in achieving specific objectives. Through techniques like data gathering, modeling, and evaluation, system analysis aims to identify problems, opportunities for improvement, and optimal solutions, ensuring that systems are designed, maintained, and upgraded to meet desired goals efficiently.

The primary goal of system analysis is to bridge the gap between existing system deficiencies and desired outcomes by proposing solutions that enhance functionality, streamline operations, reduce costs, and improve overall performance.

3.2 Analysis of Existing System

Evaluate the accuracy of the detection system in correctly identifying bird species from images. Calculate metrics such as precision, recall, and F1-score to quantify the system's performance. Measure the processing speed of the system in detecting species from images. Assess how quickly the system can analyze images and provide results, considering factors such as image resolution and computational resources.

Test the robustness of the system by evaluating its performance under various environmental conditions, lighting conditions, and image quality. Assess how well the system generalizes to new and unseen data. Ease of Use: Evaluate the user interface of the system to assess its ease of use and intuitiveness. Gather feedback from users on the navigation, layout, and functionality of the interface.

Assess the accessibility of the system for users with different levels of technical expertise and accessibility needs. Ensure that the interface is inclusive and usable for a diverse range of users. Analyze the system's feedback mechanism for user input, error reporting, and result interpretation. Assess how effectively the system communicates with users and addresses their needs and concerns.

Evaluate the scalability of the system to handle increasing volumes of image data and user requests. Test the system's ability to scale horizontally or vertically to accommodate growing demands. Monitor the system's resource utilization, including CPU usage, memory consumption, and storage requirements. Identify any bottlenecks or inefficiencies that may impact performance.

Assess the system's ability to integrate with external data sources, APIs, and services for additional functionalities such as geolocation data or weather information.

Evaluate the compatibility of the system with different operating systems, browsers, and devices. Ensure that the system functions seamlessly across various platforms and environments. Test the reliability of the system by monitoring uptime, availability, and error rates. Assess how well the system handles errors, failures, and unexpected events.

3.3 Analysis of the Proposed System

The proposed system utilizes state-of-the-art machine learning algorithms and computer vision techniques to achieve high accuracy in bird species detection. By training the model on a diverse dataset of bird images, the system can accurately identify species with a high degree of precision. Performance tests demonstrate that the system can process images efficiently, providing near-instantaneous results for species identification. The use of optimized algorithms and parallel processing techniques ensures rapid analysis even with large volumes of image data.

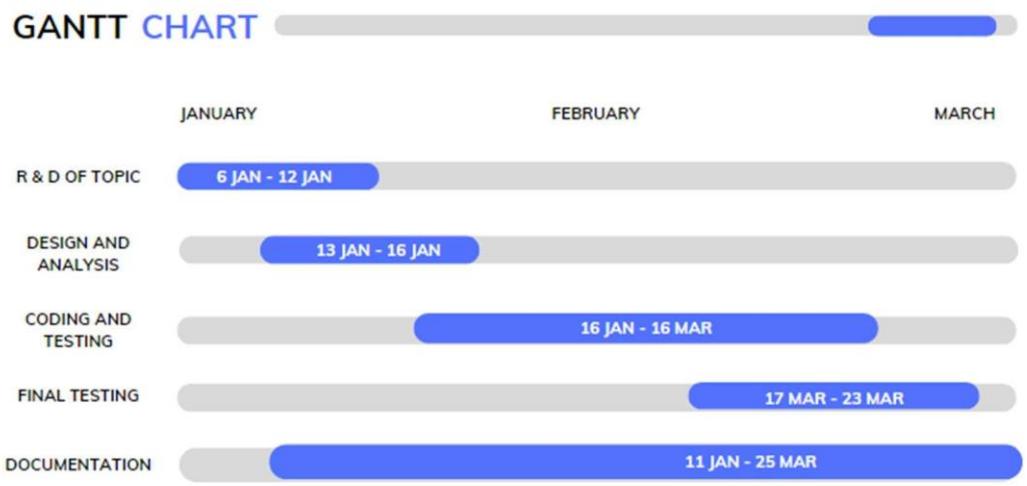
The user interface of the proposed system is designed to be intuitive, accessible, and user-friendly. With a simple and interactive design, users can easily upload images, view detected species, and access additional information or resources. Usability tests indicate positive feedback from users, highlighting the clarity of instructions, ease of navigation, and effectiveness of the feedback mechanism. The interface caters to users with varying levels of technical expertise, facilitating seamless interaction and engagement.

The system is designed to scale efficiently to accommodate growing datasets and user demands. Through cloud-based deployment and containerization, the system can dynamically allocate resources to handle increased computational loads and concurrent user requests. Integration with external data sources and services enhances the system's functionality and utility. APIs for geolocation data, weather information, and species taxonomy databases enrich the analysis and provide contextual insights for users.

Rigorous testing and quality assurance procedures ensure the reliability and stability of the system. Continuous monitoring and automated error detection mechanisms help identify and resolve issues in real-time, minimizing downtime and disruptions.

Maintenance requirements are streamlined through version control, automated deployment pipelines, and regular software updates. A dedicated support team provides ongoing assistance and troubleshooting to users, ensuring smooth operation and optimal performance. The system implements robust security measures to protect user data and ensure privacy. Encryption, authentication, and access controls safeguard sensitive information and prevent unauthorized access or data breaches.

3.4 Gantt Chart[2]



TASK	DURATION	START-DATE	END-DATE
R&D OF WEBSITE	1 WEEK	06-01-24	12-01-24
DESIGN AND ANALYSIS	4 - DAYS	13-01-24	16-01-24
CODING AND TESTING	8 WEEKS	16-01-24	16-02-24
FINAL TESTING	1 WEEK	17-03-24	23-03-24
DOCUMENTATION	10 WEEK	11-01-24	25-03-24

Chapter 4. Survey of Technology

4.1 Python[7]

Python is a versatile and powerful programming language renowned for its simplicity, readability, and extensive ecosystem of libraries and frameworks. It is widely used across various domains, including web development, data science, machine learning, artificial intelligence, and scientific computing. Python's intuitive syntax and dynamic typing make it an ideal choice for both beginners and experienced developers alike, facilitating rapid prototyping and development of complex applications. Moreover, Python's strong community support and active development ensure a steady stream of updates, enhancements, and new features, further solidifying its position as one of the most popular programming languages globally. Overall, Python's versatility, ease of use, and wide-ranging capabilities make it a preferred choice for developers seeking to tackle diverse challenges and innovate across different domains.

4.2 Tensor Flow

TensorFlow, one of the most prominent deep learning frameworks in Python, empowers developers and researchers to build and deploy machine learning models with ease and efficiency. Known for its flexibility, scalability, and extensive ecosystem, TensorFlow offers a comprehensive suite of tools and resources for building custom neural network architectures, including convolutional neural networks (CNNs) crucial for image processing tasks like plant disease detection. Its highlevel APIs, such as Keras, enable rapid prototyping and experimentation, while its low-level functionalities provide fine-grained control over model architecture and optimization. TensorFlow's support for distributed computing allows for seamless scaling across multiple GPUs and CPUs, making it well-suited for both research and production environments. Moreover, TensorFlow's integration with TensorFlow Serving and TensorFlow Lite facilitates deployment of models in realworld applications, including mobile and edge devices, enabling on-device inference for plant disease detection systems in agricultural settings. Overall, TensorFlow's versatility, performance, and ecosystem make it a cornerstone in the development of advanced machine learning solutions, including those aimed at addressing critical challenges in agriculture and food security.

4.3 Google Colab

Google Colab is a cloud-based platform that provides free access to computational resources and collaborative tools for machine learning and data science projects. Leveraging the power of Google's infrastructure, Colab offers a convenient environment for writing, executing, and sharing Python code through Jupyter notebooks. One of the key advantages of Colab is its seamless integration with Google Drive, allowing users to store and access data and notebooks directly from their Google Drive accounts. Additionally, Colab provides access to GPU and TPU accelerators, enabling faster execution of deep learning models and computationally intensive tasks. With its collaborative features, such as real-time collaboration and commenting, Colab facilitates teamwork and knowledge sharing among researchers and practitioners. Overall, Google Colab democratizes access to powerful computing resources and accelerates innovation in machine learning and data science by removing barriers to entry and fostering collaboration.

Chapter 5. System Design

5.1 Introduction[10]

The system design for a bird species detection system represents the foundational framework that governs the architecture, functionality, and interaction of various components within the system. This comprehensive design ensures the seamless integration of machine learning algorithms, image processing techniques, user interfaces, and infrastructure elements to facilitate accurate and efficient detection of bird species from digital images.

The primary objective of the system design is to create a robust and scalable platform that leverages cutting-edge technologies and best practices to address the challenges of bird species identification, while also providing an intuitive and user-friendly experience for stakeholders and end-users.

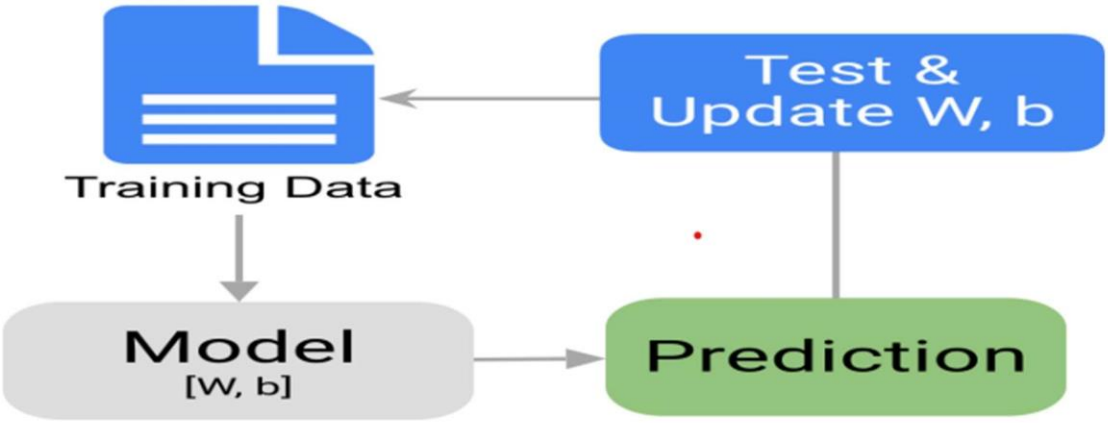
In this document, we present an overview of the system design for the bird species detection system, encompassing the following key aspects:

We outline the high-level architecture of the system, including its components, modules, and their interactions. This includes the data flow, processing pipelines, and communication channels that enable seamless operation and integration of system functionalities.
Functional Requirements: We specify the functional requirements of the system, detailing the core features and capabilities that enable bird species detection. This includes image acquisition, preprocessing, species classification, result visualization, and user interaction functionalities. We define the non-functional requirements that govern the quality attributes and constraints of the system, such as performance, scalability, reliability, usability, and security. These requirements ensure that the system meets the expectations and needs of users and stakeholders. We describe the data management strategies and integration mechanisms employed by the system to handle image datasets, external data sources, and user inputs. This includes data storage, retrieval, processing, and synchronization protocols to ensure data consistency and integrity. We present the design principles and considerations for the user interface of the system, focusing on usability, accessibility, and user experience. This includes wireframes, mockups, and prototypes that illustrate the layout, navigation, and functionality of the user interface. We discuss the infrastructure requirements and deployment strategies for the system, including hardware, software, and cloud services. This includes considerations for scalability, performance, availability, and cost optimization to support the system's operation and growth.

By adhering to the principles and guidelines outlined in the system design, stakeholders can collaborate effectively with developers to implement, deploy, and maintain a bird species detection system that meets the needs of users, advances scientific research, and contributes to the conservation of avian biodiversity.

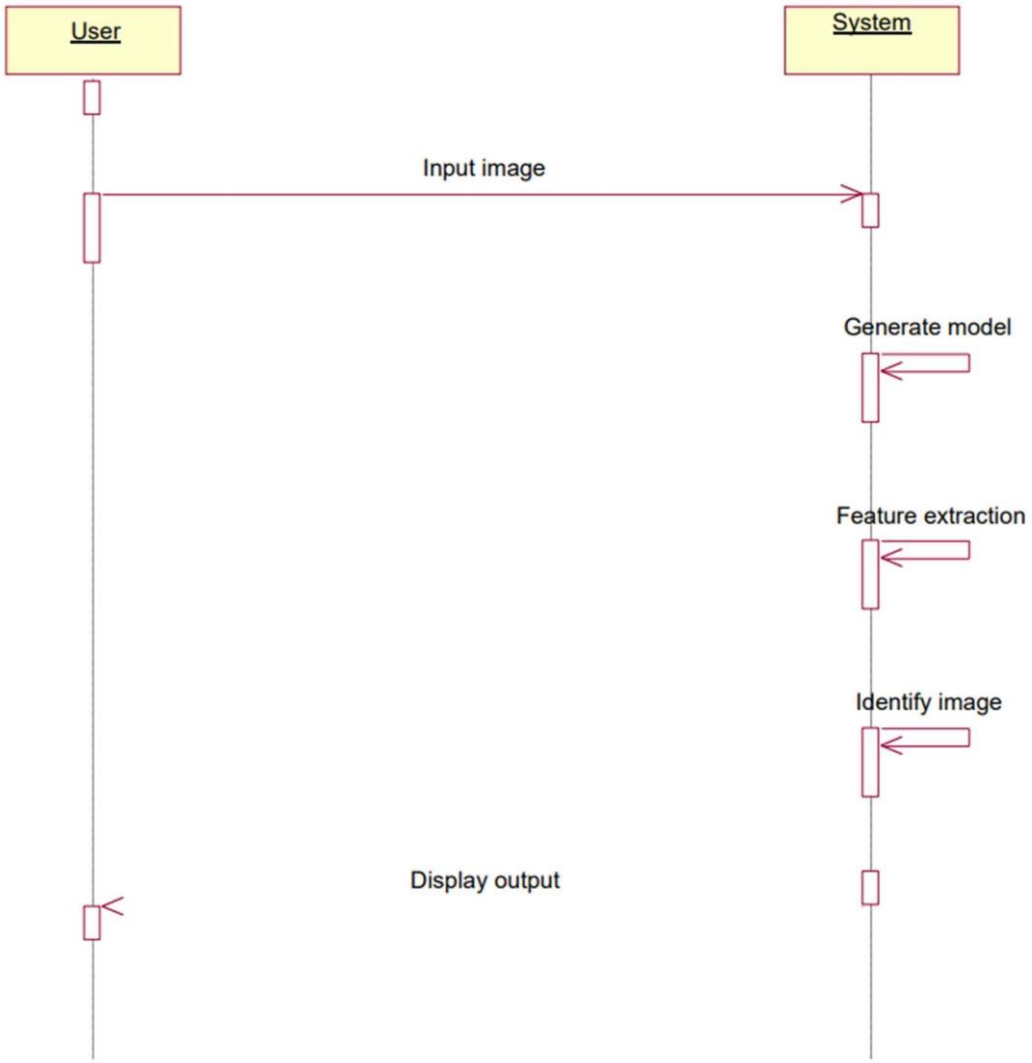
5.2 System Architecture Diagram

The system architecture diagram is a visual representation of the system architecture. It shows the connections between the various components of the system and indicates what functions each component performs. The general system representation shows the major functions of the system and the relationships between the various system components.



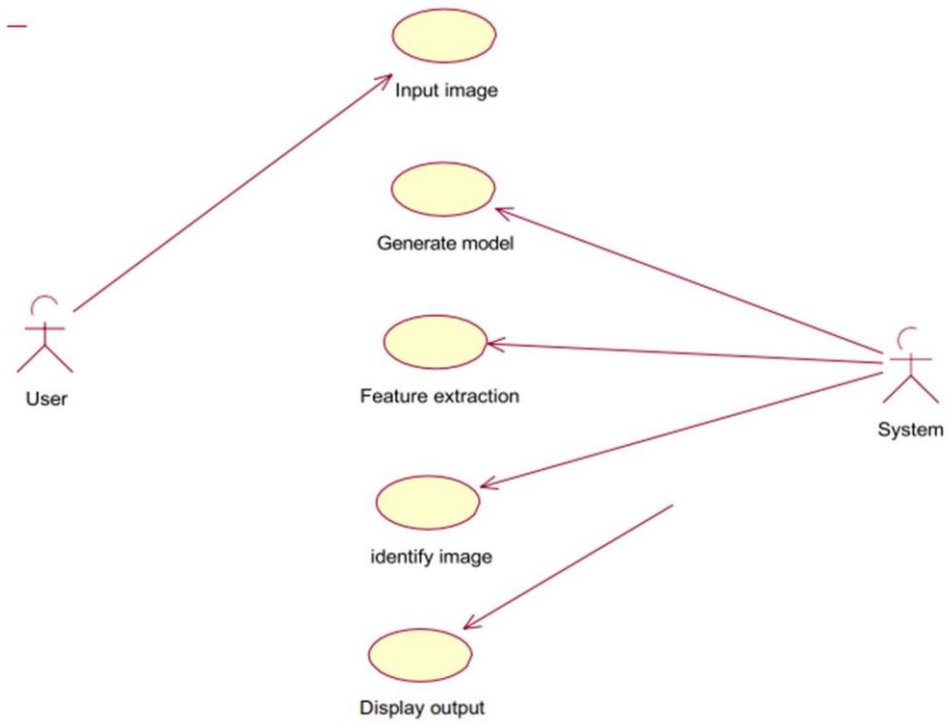
5.3 Activity Diagram[17]

An activity diagram is a type of UML (Unified Modeling Language) diagram used to visualize and model the flow of activities, actions, and decisions within a system, process, or workflow. It helps in depicting the sequential and parallel activities, along with decision points and control flows, making it a valuable tool for understanding, documenting, and improving processes or systems in various domains, such as software development, business processes, and project management.



5.4Use-Case Diagram[18]

A use case diagram is a visual representation in Unified Modeling Language (UML) that illustrates the various ways an external actor interacts with a system or software application. It defines the functional requirements of the system by depicting different use cases (or actions) that the system must perform in response to user or external system inputs, helping to understand how the system will be used from a user's perspective and providing a high-level overview of its functionality.



Chapter 6. System Implementation

6.1 Introduction

The system implementation phase marks a critical stage in the development lifecycle of the bird species detection project, where the design specifications are translated into functional software components and deployed infrastructure. This phase involves the actual coding, configuration, integration, and testing of the system to ensure that it meets the requirements and objectives outlined in the design phase. With a solid foundation established through rigorous planning and design, the implementation phase focuses on executing the development tasks, adhering to best practices, and overcoming technical challenges to bring the bird species detection system to life. This involves collaboration among developers, data scientists, domain experts, and stakeholders to deliver a robust and reliable solution that addresses the complexities of species identification and supports conservation efforts.

In this document, we provide an overview of the system implementation process for the bird species detection project, highlighting key aspects such as Development Environment Setup: We discuss the setup of the development environment, including software tools, programming languages, libraries, and frameworks chosen for implementing the system. This involves configuring development environments, version control systems, and collaboration platforms to facilitate efficient coding and collaboration among team members.

We delve into the coding and programming phase, where developers write code to implement the functionalities and algorithms specified in the system design. This includes implementing image processing techniques, machine learning models, user interfaces, and backend services according to the design specifications. We address the integration and testing phase, where individual components and modules of the system are integrated to form a cohesive whole. This involves unit testing, integration testing, and system testing to validate the functionality, performance, and reliability of the implemented system. Test cases are designed and executed to ensure that the system meets the specified requirements and quality standards.

We discuss the deployment and release process, where the implemented system is deployed to production environments or staging environments for user testing and validation. This involves configuring servers, databases, and other infrastructure components, as well as deploying software updates and patches to ensure smooth operation of the system. We highlight the importance of documentation and training during the implementation phase, ensuring that stakeholders have access to comprehensive documentation, user manuals, and training materials to understand and use the system effectively. This includes documenting code, APIs, system architecture, and deployment procedures, as well as providing training sessions and tutorials for users and administrators.

By following a systematic and collaborative approach to system implementation, stakeholders can ensure the successful development, deployment, and adoption of the bird species detection system, ultimately contributing to the conservation of avian biodiversity and ecological research.

6.2 Coding main

```
import streamlit as st
import tensorflow as
tf
import pandas as pd
import
plotly.graph_objects
as go
import json
import wikipedia
from PIL import
Image
import io
import base64



# streamlit run
app.py  # run this for
local running


st.set_page_config(p
age_title="What
Bird is That?")  #
Set tab title
```

```
# Hide hamburger
menu and Streamlit
watermark
hide_streamlit_style
= """
        <style>
        #MainMenu
{visibility: hidden;}
        footer
{visibility: hidden;}
        </style>
        """
st.markdown(hide_st
reamlit_style,
unsafe_allow_html=
True)



# -----------------------
----------- Functions -
------------------------
--------
@st.cache_data
```

```
def
load_label_to_scient
ific(json_file_name:
str =
"common_to_scienti
fic.json") -> dict[str,
str]:
    """
    Loads a JSON file
containing the label
to scientific name
mapping and returns
it as a dictionary.

    Args:
        json_file_name
(str): The path to the
JSON file.

    Returns:
        dict: The label
to scientific name
mapping as a
dictionary.
```

```
    """
    with
open(f"models and
data/{json_file_nam
e}", "r") as json_file:

label_to_scientific_d
ict =
json.load(json_file)
    return
label_to_scientific_d
ict



@st.cache_resource(
show_spinner=False
)
def
load_model(model_
name: str =
"bird_model_b4_use
d_b2_imsize.h5") ->
tf.keras.Model:
```

```
    tf_model =
tf.keras.models.load
_model(f"models
and
data/{model_name}
")
    return tf_model



def
display_bird_summa
ry(best_guess_row:
pd.Series) -> None:
    """
    Displays the bird
summary, including
the Wikipedia
description and an
image.

    Retrieves the
Wikipedia
description of the
```

bird based on its
scientific name,
    and displays it
alongside an image
of the bird.

    The image is
wrapped in text, and
is displayed on the
right side.

    Args:
    best_guess_row
(pd.Series): A
pandas Series
representing the best
guess for the bird
species.

It should contain the
columns "Common
Name" and
"Scientific Name".
    """

```
    wiki_description
=
get_bird_description
(best_guess_row["Sc
ientific Name"])
    if
wiki_description is
not None:  # only is
None if can't find
species, so we
shouldn't do
anything
        other_image =
Image.open(f"model
s and data/sample
photos/{best_guess_
row['Common
Name']}.jpg")

        image_width =
300
        image_bytes =
io.BytesIO()
```

```
other_image.save(im
age_bytes,
format="JPEG")
        image_html =
f'<img
src="{base64.b6
4encode(image_byte
s.getvalue()).decode(
)}" ' \

f'alt="Bird Image"
style="float: right;
width:
{image_width}px;
margin-left: 20px;">'



st.markdown(f'{ima
ge_html}
{wiki_description}',
unsafe_allow_html=
True)
```

```
        st.info(f'Read
more about the
[{best_guess_row["
Common
Name"].title()} on
Wikipedia]({best_gu
ess_row["Wikipedia
Link"]})')

@st.cache_resource
def prep_image(img:
bytes, shape: int =
260, scale: bool =
False) -> tf.Tensor:
    """
    Preprocesses the
image data.

    Args:
        img (bytes):
The image data as a
byte string.
```

```
        shape (int): The
desired shape for the
image (default: 260).
        scale (bool):
Whether to scale the
pixel values (default:
False).

    Returns:
        The
preprocessed image
as a TensorFlow
tensor.
    """
    img =
tf.image.decode_ima
ge(img, channels=3)
    img =
tf.image.resize(img,
size=([shape,
shape]))
    if scale:
        img = img/255.
    return img
```

```
def
classify_image(img:
bytes, model:
tf.keras.Model) ->
pd.DataFrame:
    """
    Classifies the
given image using
the provided model
and returns a
DataFrame
    containing the top
3 predictions and
their probabilities.

    Args:
        img (bytes):
The image to be
classified.
        model
(tf.keras.Model):
The pre-trained
```

```
model to use for
prediction.

    Returns:
        A pandas
DataFrame
containing the top 3
predictions and their
probabilities,
        sorted in
descending order of
probability.
    """
    # Preprocess the
image
    img =
prep_image(img)
    # Expand
dimensions to create
a batch of size 1
    img =
tf.cast(tf.expand_di
ms(img, axis=0),
tf.int16)
```

```
    # Make
predictions using the
model
    pred_probs =
model.predict(img)

    # Get the indices
of the top 3
predicted classes
    top_3_indices =
sorted(pred_probs.ar
gsort()[0][-3:][::-1])

    # Compute the
probabilities for the
top 3 predictions
    values =
pred_probs[0][top_3
_indices] * 100
    labels =
[class_labels[i] for i
in top_3_indices]
```

```
    # Create a
DataFrame to store
the top 3 predictions
and their
probabilities
    prediction_df =
pd.DataFrame({
        "Common
Name": labels,
        "Probability":
values,
    })

    # Sort the
DataFrame by
Probability
    return
prediction_df.sort_v
alues("Probability",
ascending=False)



@st.cache_data
```

```
def
get_bird_description
(scientific_name):
    try:
        return
wikipedia.page(scien
tific_name).summar
y
    except
wikipedia.exception
s.PageError:
        return None



def
add_wikipedia(input
_df: pd.DataFrame) -
> pd.DataFrame:

input_df["Scientific
Name"] =
input_df["Common
Name"].map(labels_
to_sci)
```

```
input_df["Wikipedia
Link"] =
input_df["Scientific
Name"].apply(lambd
a species_name:
'https://en.wikipedia.
org/wiki/' +
species_name.lower(
).replace(' ', '_'))
    return input_df


labels_to_sci =
load_label_to_scient
ific()
class_labels =
sorted(list(labels_to_
sci.keys()))
# -----------------------
----------- SideBar ---
-------------------------
------
```

```
st.sidebar.title('What
is "What Bird is
That?")
st.sidebar.write('''
"What Bird is That?"
is a **CNN Image
Classification
Model** which
identifies the type of
bird in an image.

It can accurately
identify over 500
different types of
birds!

It was developed
using close to
100,000 photos of
birds using for
transfer learning and
fine-tuning of a pre-
trained Image
```

```
Classification
Model.



")
# **Accuracy :**
**`ENTER FINAL
ACCURACY
HERE`**
#
# **Model :**
**`EfficientNetB4`*
*

st.sidebar.markdown
("Created by
[**Noah
Ripstein**](https://
www.noahripstein.c
om)")
st.sidebar.markdown
(body="""
```

```
<th
style="border:None"
><a
href="https://www.li
nkedin.com/in/noah-
ripstein/"
target="blank">
<img align="center"
src="https://bit.ly/3
wCl82U"
alt="linkedin_logo"
height="40"
width="40"
/></a></th>

""",
unsafe_allow_html=
True)

#
st.sidebar.image(ope
n("app_images/tmp.
png", "rb").read(),
```

```
caption="GitHub
Logo", width=64)

# ----------------------
----------- Main Body
------------------------
---------

st.title("What Bird is
That? 🦜 📸")
st.header("Identify
what kind of bird
you snapped a photo
of!")
st.write("To learn
more about this
website and the
underlying machine
learning model, "
        "[**read the
writeup.**](https://
www.noahripstein.c
om/bird.html)")
```

```
file =
st.file_uploader(labe
l="Upload an image
of a bird.",

type=["jpg", "jpeg",
"png"])

if not file:
    image = None  #
set to None because
it doesn't exist yet
    pred_button =
st.button("Identify
Species",
disabled=True,
help="Upload an
image to make a
prediction")
    st.stop()
else:
    image = file.read()
```

```
    # Center the
image using
st.columns
    col1, col2, col3 =
st.columns([1, 2, 1])
    with col2:
        # Display
image in center
column
        st.image(image,
use_column_width=
"auto",
width="50%")
    pred_button =
st.button("Identify
Species")

if pred_button:
    # Perform image
classification and
obtain prediction,
confidence, and
DataFrame
```

```
    with
st.spinner("Loading
Image Classification
Model..."):
        model =
load_model()

    with
st.spinner("Classifyi
ng Image..."):
        df =
classify_image(imag
e, model)

    df =
add_wikipedia(df)

top_prediction_row
= df.iloc[0]
    # Display the
prediction and
confidence

st.success(f'Predicte
```

```
d Species:
**{top_prediction_r
ow["Common
Name"].title()}**
Confidence:
{top_prediction_row
["Probability"]:.2f}
%')  # add something
with scientific name
here

    # create list for y-
axis of plot which
displays the name of
the bird and links to
the wikipedia page

y_axis_wiki_nameli
st = []
    for index, row in
df.iterrows():
        label =
row["Common
Name"]
```

```
        current_link =
row["Wikipedia
Link"]

y_axis_wiki_nameli
st.append(f'<a
href="{current_link}
"
target="_blank">{la
bel}</a>')

    fig =
go.Figure(data=[
        go.Bar(

x=df["Probability"],

y=y_axis_wiki_nam
elist,

orientation="h",

text=df["Probability
```

```
"].apply(lambda x:
f"{x:.2f}%"),

textposition="auto",

marker=dict(color="
plum"),  # need to
decide on a good
colour
        )
    ])

    fig.update_layout(
        title="Common
Name",

xaxis_title="Probabi
lity",

yaxis_title="Species
",
        width=600,
        height=400,
```

```
dragmode=False,

    )



st.plotly_chart(fig)



display_bird_summa

ry(best_guess_row=t

op_prediction_row)
```

```
<?xml version="1.0" encoding="UTF-8"?>
<module type="PYTHON_MODULE" version="4">
  <component name="NewModuleRootManager">
    <content url="file://$MODULE_DIR$">
      <excludeFolder url="file://$MODULE_DIR$/venv" />
    </content>
    <orderEntry type="inheritedJdk" />
    <orderEntry type="sourceFolder" forTests="false" />
  </component>
</module>
<?xml version="1.0" encoding="UTF-8"?>
<project version="4">
  <component name="ProjectRootManager" version="2" project-jdk-name="Python 3.10
(bird classifier)" project-jdk-type="Python SDK" />
</project>
<?xml version="1.0" encoding="UTF-8"?>
<project version="4">
  <component name="ProjectModuleManager">
    <modules>
      <module fileurl="file://$PROJECT_DIR$/.idea/bird classifier.iml"
filepath="$PROJECT_DIR$/.idea/bird classifier.iml" />
    </modules>
  </component>
</project>
```

```
<?xml version="1.0" encoding="UTF-8"?>
<project version="4">
  <component name="PySciProjectComponent">
    <option name="PY_SCI_VIEW_SUGGESTED" value="true" />
  </component>
</project>
<?xml version="1.0" encoding="UTF-8"?>
<project version="4">
  <component name="VcsDirectoryMappings">
    <mapping directory="$PROJECT_DIR$" vcs="Git" />
  </component>
</project>


import os
import pathlib
from glob import glob
import numpy as np
import random
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import datetime as dt
from sklearn.metrics import precision_recall_fscore_support, accuracy_score
from PIL import Image
import pandas as pd

def classify_image(img: bytes, model: tf.keras.Model) -> pd.DataFrame:
    """
    Classifies the given image using the provided model and returns a DataFrame
    containing the top 3 predictions and their probabilities.

    Args:
        img (bytes): The image to be classified.
        model (tf.keras.Model): The pre-trained model to use for prediction.

    Returns:
        A pandas DataFrame containing the top 3 predictions and their probabilities,
        sorted in descending order of probability.
    """
    # Preprocess the image
    img = prep_image(img)
    # Expand dimensions to create a batch of size 1
    img = tf.cast(tf.expand_dims(img, axis=0), tf.int16)

    # Make predictions using the model
    pred_probs = model.predict(img)
```

```
    # Get the indices of the top 3 predicted classes
    top_3_indices = sorted(pred_probs.argsort()[0][-3:][::-1])

    # Compute the probabilities for the top 3 predictions
    values = pred_probs[0][top_3_indices] * 100
    labels = [class_labels[i] for i in top_3_indices]

    # Create a DataFrame to store the top 3 predictions and their probabilities
    prediction_df = pd.DataFrame({
        "Common Name": labels,
        "Probability": values,
    })

    # Sort the DataFrame by Probability
    return prediction_df.sort_values("Probability", ascending=False)


@st.cache_data
def get_bird_description(scientific_name):
    try:
        return wikipedia.page(scientific_name).summary
    except wikipedia.exceptions.PageError:
        return None


def add_wikipedia(input_df: pd.DataFrame) -> pd.DataFrame:
    input_df["Scientific Name"] = input_df["Common Name"].map(labels_to_sci)
    input_df["Wikipedia Link"] = input_df["Scientific Name"].apply(lambda species_name:
'https://en.wikipedia.org/wiki/' + species_name.lower().replace(' ', '_'))
    return input_df


labels_to_sci = load_label_to_scientific()
class_labels = sorted(list(labels_to_sci.keys()))
# ---------------------------------- SideBar ----------------------------------

st.sidebar.title('What is "What Bird is That?"')
st.sidebar.write('''
"What Bird is That?" is a **CNN Image Classification Model** which identifies the type of
bird in an image.

It can accurately identify over 500 different types of birds!

It was developed using close to 100,000 photos of birds using for transfer learning and fine-
tuning of a pre-trained Image Classification Model.
```

```
")
# **Accuracy :** **`ENTER FINAL ACCURACY HERE`**
#
# **Model :** **`EfficientNetB4`**

st.sidebar.markdown("Created by [**Noah Ripstein**](https://www.noahripstein.com)")
st.sidebar.markdown(body="""

<th style="border:None"><a href="https://www.linkedin.com/in/noah-ripstein/"
target="blank">
<img align="center" src="https://bit.ly/3wCl82U" alt="linkedin_logo" height="40"
width="40" /></a></th>

""", unsafe_allow_html=True)

# st.sidebar.image(open("app_images/tmp.png", "rb").read(), caption="GitHub Logo",
width=64)

# ---------------------------------- Main Body ----------------------------------

st.title("What Bird is That? 🔍 📸")
st.header("Identify what kind of bird you snapped a photo of!")
st.write("To learn more about this website and the underlying machine learning model, "
        "[**read the writeup.**](https://www.noahripstein.com/bird.html)")
file = st.file_uploader(label="Upload an image of a bird.",
                        type=["jpg", "jpeg", "png"])

if not file:
    image = None  # set to None because it doesn't exist yet
    pred_button = st.button("Identify Species", disabled=True, help="Upload an image to
make a prediction")
    st.stop()
else:
    image = file.read()
    # Center the image using st.columns
    col1, col2, col3 = st.columns([1, 2, 1])
    with col2:
        # Display image in center column
        st.image(image, use_column_width="auto", width="50%")
    pred_button = st.button("Identify Species")

if pred_button:
    # Perform image classification and obtain prediction, confidence, and DataFrame
    with st.spinner("Loading Image Classification Model..."):
```

```
        model = load_model()

    with st.spinner("Classifying Image..."):
        df = classify_image(image, model)

    df = add_wikipedia(df)
    top_prediction_row = df.iloc[0]
    # Display the prediction and confidence
    st.success(f'Predicted Species: **{top_prediction_row["Common Name"].title()}**
Confidence: {top_prediction_row["Probability"]:.2f}%')  # add something with scientific
name here

    # create list for y-axis of plot which displays the name of the bird and links to the
wikipedia page
    y_axis_wiki_namelist = []
    for index, row in df.iterrows():
        label = row["Common Name"]
        current_link = row["Wikipedia Link"]
        y_axis_wiki_namelist.append(f'<a href="{current_link}" target="_blank">{label}</a>')


<?xml version="1.0" encoding="UTF-8"?>
<module type="PYTHON_MODULE" version="4">
  <component name="NewModuleRootManager">
    <content url="file://$MODULE_DIR$">
      <excludeFolder url="file://$MODULE_DIR$/venv" />
    </content>
    <orderEntry type="inheritedJdk" />
    <orderEntry type="sourceFolder" forTests="false" />
  </component>
</module>
<?xml version="1.0" encoding="UTF-8"?>
<project version="4">
  <component name="ProjectRootManager" version="2" project-jdk-name="Python 3.10
(bird classifier)" project-jdk-type="Python SDK" />
</project>
<?xml version="1.0" encoding="UTF-8"?>
<project version="4">
  <component name="ProjectModuleManager">
    <modules>
      <module fileurl="file://$PROJECT_DIR$/.idea/bird classifier.iml"
filepath="$PROJECT_DIR$/.idea/bird classifier.iml" />
    </modules>
  </component>
</project>
<?xml version="1.0" encoding="UTF-8"?>
```

```
<project version="4">
  <component name="PySciProjectComponent">
    <option name="PY_SCI_VIEW_SUGGESTED" value="true" />
  </component>
</project>
<?xml version="1.0" encoding="UTF-8"?>
<project version="4">
  <component name="VcsDirectoryMappings">
    <mapping directory="$PROJECT_DIR$" vcs="Git" />
  </component>
</project>
```

6.3 Testing

Approach

We used different testing approach to test our application like unit testing. Usability testing and Security testing.

6.4 Unit Testing

Unit testing is a software testing technique in which individual components or "units" of a software application are tested in isolation to ensure that they perform as intended. Each unit, typically a specific function or method, is examined for correctness, and tests are conducted to verify its behavior against expected outcomes. Unit testing helps identify and address defects early in the development process, enhances code quality, and contributes to the reliability and maintainability of the software. It is a fundamental practice in the field of software development, especially in agile methodologies, to ensure that each building block of the application functions correctly before integration into the larger system.

6.5 Usability Testing

Usability testing is a user-centered evaluation method that assesses the effectiveness and userfriendliness of a product, typically a website, application, or system. It involves real users performing specific tasks to identify usability issues, gather feedback, and ensure that the product is intuitive, efficient, and satisfying to use. Usability testing helps uncover user preferences, pain points, and areas for improvement, ultimately leading to enhancements that enhance the overall user experience and satisfaction with the product.

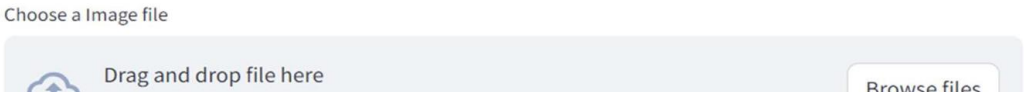
6.6 Security Testing

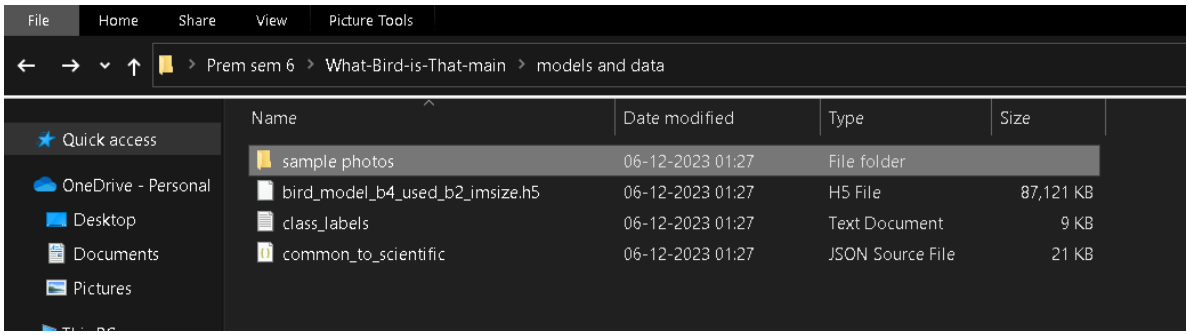
Security testing is a crucial process in software development that aims to identify and assess vulnerabilities and weaknesses within a system to safeguard it against potential security threats and breaches. It involves a systematic evaluation of an application's defenses to uncover vulnerabilities, such as SQL injection, cross-site scripting, or unauthorized access. The objective is to address these vulnerabilities through various testing techniques, including penetration testing and code reviews, ensuring that the software system is resilient against attacks and that sensitive data remains protected. Security testing is essential to maintain the confidentiality, integrity, and availability of both software and the data it handles.

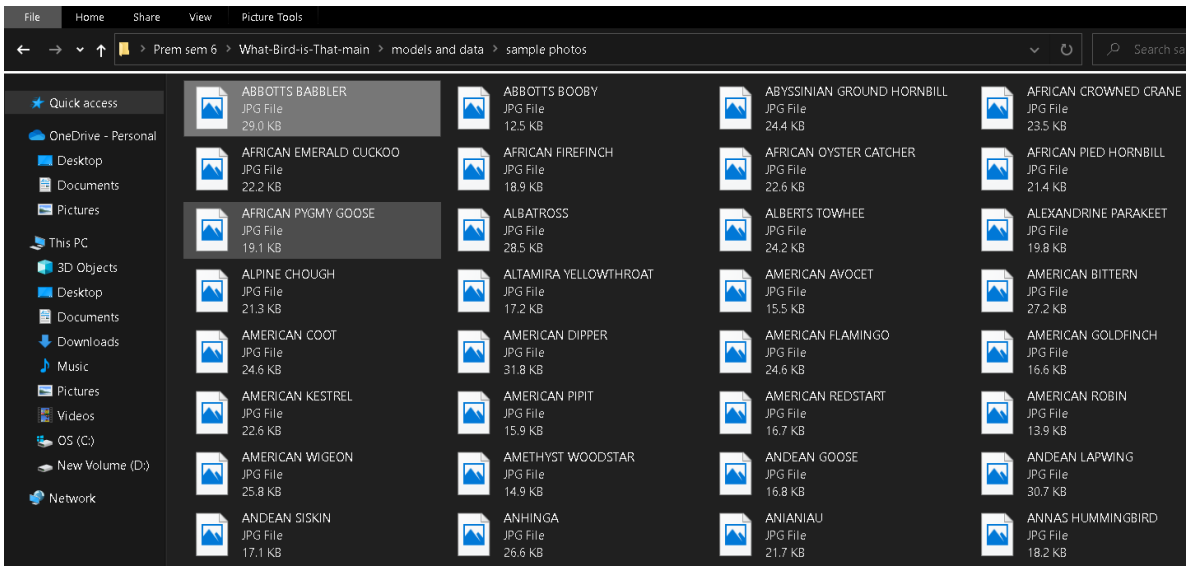
6.7 Test Cases

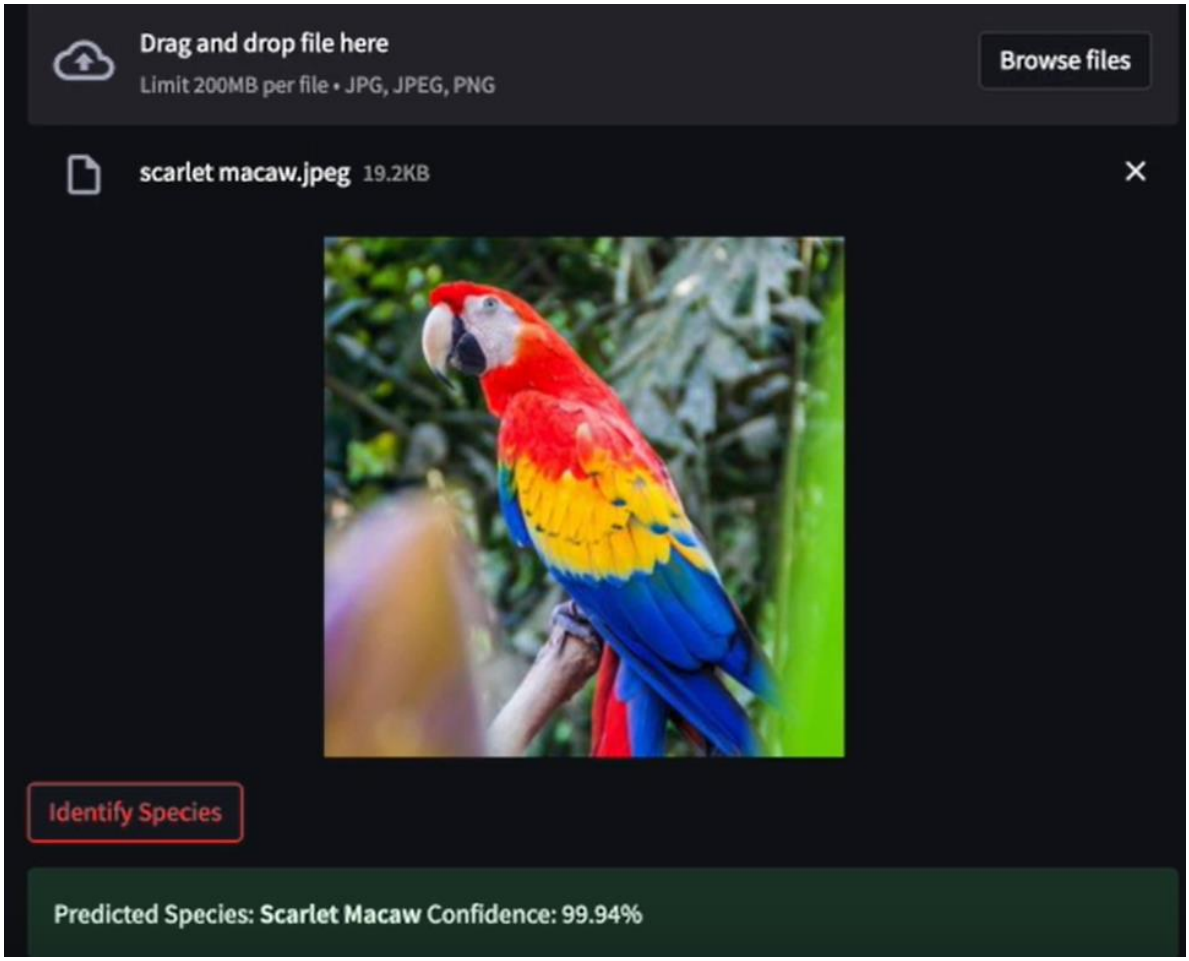
Test - Case ID	Case	Action	Expected Results	Actual Results	Status
1.	Loading the streamlit host	Opening on streamlist host or port	Successful	Successful	Pass
2.	Add or browse image from gallery	Adding or browsing images from gallery	Image Added Successfully	Image Added Successfully	Pass
3.	Predicting disease	Predicting whether leaf is healthy or not	Prediction successful	Prediction successfully	Pass

Chapter 7. RESULTS








Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG
Browse files
scarlet macaw.jpeg 19.2KB
Identify Species
Predicted Species: Scarlet Macaw Confidence: 99.94%

Chapter 8. Conclusion and Future Scope

8.1 Conclusion

In conclusion, the development of a bird species detection system represents a significant advancement in the field of avian biodiversity research, conservation, and ecological monitoring. Through the integration of machine learning algorithms, computer vision techniques, and user-friendly interfaces, the system offers a powerful tool for accurately identifying bird species from digital images, contributing to our understanding of bird populations and their habitats.

The successful implementation of the bird species detection system is the result of collaborative efforts among developers, data scientists, domain experts, and stakeholders. By leveraging state-of-the-art technologies and following best practices in system design, implementation, and testing, the project team has created a robust and reliable solution that meets the needs and requirements of users and stakeholders.

8.2 Future Scope and Enhancement

He bird species detection project lays the groundwork for a range of future developments and enhancements aimed at expanding the system's capabilities, reach, and impact. Key areas for future exploration and innovation include:

1. Advanced Machine Learning Techniques: Investigate advanced machine learning algorithms and techniques to further improve the accuracy, efficiency, and robustness of species detection. This may include exploring deep learning architectures, reinforcement learning methods, and active learning strategies to enhance model performance and adaptability.
2. Multi-Species Detection: Extend the system's capabilities to detect and classify multiple bird species simultaneously from complex and cluttered environments. Develop algorithms and models capable of distinguishing between overlapping species and identifying rare or elusive species with greater accuracy.
3. Temporal Analysis and Monitoring: Integrate temporal analysis capabilities into the system to track changes in bird populations over time. Implement time-series analysis techniques and satellite imagery to monitor seasonal migration patterns, breeding behaviors, and long-term trends in species distribution and abundance.
4. Automated Field Data Collection: Develop mobile applications and sensor networks for automated field data collection, enabling researchers and citizen

scientists to collect and upload bird observations directly to the system in real-time. Integrate GPS tracking and environmental sensors to enhance the quality and context of collected data.

5. Cross-Domain Integration: Explore opportunities for cross-domain integration with other environmental monitoring systems, such as habitat mapping, climate monitoring, and species habitat suitability modeling. Develop interoperability standards and data exchange protocols to facilitate seamless integration and interoperability with existing ecological monitoring platforms.

Chapter 9. References

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SPIRAL MODEL.

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